

Timothy James Wood

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SUMMARY

I have completed two research projects in statistical population genetics under Prof. Mark Beaumont, covering demographic inference and genome-wide selection scans. I am seeking a PhD in statistical genetics and machine learning, with a particular interest in inference methodology for genomic data.

EDUCATION

University of Bristol Sept 2026 – Sept 2027
MSc Bioinformatics TBC

- HDR UK MSc Scholarship (charity partner: Coeliac UK)

University of Bristol Sept 2023 – Jun 2026
BSc Biological Sciences First Class Honours

- First-class marks in both the final-year research project (75%) and literature review (78%)

RESEARCH EXPERIENCE

School of Mathematics, University of Bristol Jun – Sept 2026
Research Assistant for Prof. Mark Beaumont and Dr Dennis Prangle timj-wood/wildcat_models

- Built a folded joint site-frequency spectrum from 46 whole-genome Scottish wildcats and domestic cats
- Fitted three two-population models in dadi, including an 11-parameter domestication and land-bridge model
- Implemented composite-likelihood model selection and Godambe confidence intervals from first principles
- Showed that deep parameters lie on a single likelihood ridge
- Diagnosed a boundary optimum that invalidated the best-scoring model, leading to a defensible primary result
- Ran optimisation as SLURM array jobs on Bristol's HPC; outputs form a baseline for simulation-based inference

BSc Research Project: Selection and Introgression in the European Wildcat Sept – Dec 2025
Supervisor: Prof. Mark Beaumont Data from Dr Jo Howard-McCombe, RZSS

- Processed whole-genome SNP data for 108 wild, captive and domestic cats (VCFtools, PLINK)
- Ran a pcadapt genome scan for selection; identified 1,275 outlier intervals and annotated 11 candidate genes
- Found significant GO enrichment for GABA, taurine and calcium-ion transport
- Developed PCA visualisations to separate introgression from demographic structure

BSc Literature Review: Engineering Predictability in Living Systems Sept – Dec 2025
Supervisor: Prof. Thomas Gorochowski Biocompute Lab

- Reviewed synthetic circuits, genome recoding and biological foundation models (Evo, ESM-2, ProGen)
- Argued that data-driven models capture evolutionary correlation rather than causal mechanism
- Proposed context-aware data and adaptive feedback as remedies

TECHNICAL SKILLS

Programming Python (NumPy, SciPy, pandas), R (tidyverse), Bash, Git, SLURM/HPC, LaTeX
Statistical genetics dadi, pcadapt, VCFtools, PLINK; composite likelihood, block bootstrap, PCA

OTHER EXPERIENCE

PwC LLP Jun – Jul 2025
Audit Intern London

- Supported the audit of WPP plc (FTSE 100); produced analytical working papers from large financial datasets

University of Bristol Engineering Rugby Society May 2025 – May 2026
President

- Ran club finances and Students' Union relations; secured sponsorship and event funding